

Applied Machine Learning for Business

Plaksha University — Monsoon 2026 Instructor: Nikhil George **Teaching Fellow:** Gajendra Dwivedi **Office Hours:** By appointment — 11am–5pm, Mon–Fri, Havells 402

Course Description

Students arrive with machine learning foundations and programming fluency. This course builds on that preparation across a range of business situations where their technical competency in data science is called upon — from exploratory analysis and model building, to evaluating systems, to the economics and institutional consequences of how these systems behave. At each step the questions are precise: what is the objective function optimising, where does the metric break down, what does scale change. Some topics are named for a business domain — customer analytics, search and advertising — others for an algorithmic framework — networks, recommender systems — but in each, the work happens at the same intersection: the data, the algorithm, and the context in which it operates. Assignments span conceptual questions, data analysis, and programming.

Prerequisites: Prior coursework in machine learning required. Concurrent deep learning enrollment acceptable.

Modules

1. Customer Analytics A brief introduction to dimensionality reduction (PCA), customer segmentation, churn prediction, and customer lifetime value — including the limitations of each. A recurring theme: the choices made before the algorithm runs shape what the numbers report.

Strongly Encouraged - McCarthy, “A Playful Introduction to Customer Lifetime Value” (2026) - Ascarza, “Retention Futility: Targeting High-Risk Customers Might Be Ineffective” (2018) — *Journal of Marketing Research*

Technical Reference (optional) - Abdi & Williams, “Principal Component Analysis” (2010) — personal.utdallas.edu/~herve/abdi-awPCA2010.pdf

2. Networks A node’s importance is a function of its neighbors’ importance — a recursive structure that degree centrality cannot capture. Eigenvector centrality, PageRank, community detection, and influence maximization: the algorithmic foundations of ranking and reach in a network.

Readings - Barabasi, *Network Science* (2016), Chapters 1–3 — free at networksciencebook.com - Page, Brin et al., “The PageRank Citation Ranking” (1999) — Stanford Tech Report - Haldane, “Rethinking the Financial Network” (2009) — Bank of England speech

3. Evaluating Deployed Systems A screening algorithm can be evaluated only on cases it approved — a property called selective labels. This module introduces the problem, develops a technique for estimating true model performance under reasonable structural assumptions, and examines what standard metrics like AUC are actually measuring when the evaluation sample is

not random. Bootstrap confidence intervals are introduced as a tool for quantifying uncertainty in performance estimates.

Strongly Encouraged - Fawcett, “An Introduction to ROC Analysis” (2006) — *Pattern Recognition Letters*

Background Reference - Kleinberg, Lakkaraju et al., “Human Decisions and Machine Predictions” (2018) — *QJE*; NBER working paper free - Efron & Hastie, *Computer Age Statistical Inference*, Chapter 10 — free PDF at Stanford

4. Search and Advertising Introduction to classical search and advertising. The module covers the auction mechanism that determines which ads are shown and at what price — including the generalized second-price auction. We then examine the cookie ecosystem and how it is changing. The attribution problem in advertising measurement comes next: an ad click that leads to a purchase is not straightforward to credit, and experiments are the primary tool for resolving it. The module closes with how deep learning has changed search and the frontier questions that arise when language models answer rather than rank.

Readings - Varian, “Position Auctions” (2007) — *International Economic Review*; free on Varian’s Berkeley page - Edelman, Ostrovsky & Schwarz, “Internet Advertising and the Generalized Second-Price Auction” (2007) — *AER* - EFF, “Behind the One-Way Mirror” (2019) — eff.org - Kohavi et al., “Controlled Experiments on the Web” (2009) — *Data Mining and Knowledge Discovery*

5. Recommender Systems Users and items can be represented as vectors in a shared latent space — the dot product between them predicts preference. This is the underlying idea of recommender systems, from collaborative filtering to modern deep learning approaches. The module introduces the core concept, touches upon progress in the field, and then covers practical challenges: cold start, sparsity, and the fundamental difficulty of evaluating a recommender system without deployment.

Strongly Encouraged - Netflix Technology Blog — netflixtechblog.medium.com

Background Reference - Koren, Bell & Volinsky, “Matrix Factorization Techniques for Recommender Systems” (2009) — *IEEE Computer* - Chaney, Stewart & Engelhardt, “How Algorithmic Confounding Increases Homogeneity” (2018) — arXiv:1710.11214

6. Economics of Modern AI An introduction to the economic and strategic questions raised by rapid progress in AI. The module opens with how machines learn representations — from word embeddings to recurrent networks to the transformer architecture — tracing the progression from count-based methods to the parallel, attention-based architecture that made large-scale training possible. From there, performance scales predictably with compute, data, and model size — an empirical regularity with known corrections and contested limits. The module covers the historical argument that scale beats specialization (Sutton), what the scaling laws actually claim and where they break (Kaplan, Hoffmann), and what it means that frontier AI is beginning to produce new results in mathematics and science (Bubeck et al.).

Strongly Encouraged - Sutton, “The Bitter Lesson” (2019) — incompleteideas.net - Kaplan et al., “Scaling Laws for Neural Language Models” (2020) — arXiv:2001.08361 - Bubeck et al., “Early

Science Acceleration with GPT-5” (2025) - Eric Jang, “Building AlphaGo from Scratch with Modern AI Tools” [video] — youtube.com/watch?v=X_ZVSPcZhtw

Background Reference - Wigner, “The Unreasonable Effectiveness of Mathematics in the Natural Sciences” (1960) - Hoffmann et al., “Training Compute-Optimal Large Language Models” (2022) — [arXiv:2203.15556](https://arxiv.org/abs/2203.15556) - Amodei, “Machines of Loving Grace” (2024) — darioamodei.com - Thompson et al., “The Computational Limits of Deep Learning” (2020) — [arXiv:2007.05558](https://arxiv.org/abs/2007.05558)

7. Time Permitting How AI progress is measured — and why that is hard: Goodhart’s law applied to benchmarks, contamination, and the claim that emergent capabilities are an artifact of discontinuous metrics rather than discontinuous capability. The module closes with the immediate business consequence: a two-hundred-billion-dollar advertising model built on search links is under structural threat from systems that answer rather than rank.

Assessment

Component	Weight
Team Assignments	35%
Class Tests (2)	30%
Individual Assignments	25%
Class Participation	10%

Class tests are conducted in-class and drawn from assignment questions. Attendance at or above 80% of sessions earns full participation credit; below that threshold earns zero.